SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID – SIH26117**
- **Problem Statement Title – Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work**
- **Theme – Smart Automation**
- **PS Category – Software**
- **Team ID –**
- **Team Name – Alpha-6**



SOVEREIGN INDUSTRIAL AI • BUILT FOR CONFIDENTIAL WORK

ON-PREM | AIR-GAPPED | MULTIMODAL | AGENTIC | AUDITABLE

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SOVEREIGN INDUSTRIAL AI WORKBENCH

Proposed Solution — a plant-grade, evidence-first AI workbench

CONFIDENTIALITY • CONTROL • TRACEABILITY • AUTOMATION

MULTIMODAL UNDERSTANDING

SOPs • P&IDs • manuals • PDFs • images • logs
OCR + vision + document parsing

AGENTIC EXECUTION

Plan → retrieve → reason → use tools → verify
Human approval for sensitive actions

SOVEREIGN BY DESIGN

On-prem / air-gapped • no external API dependency
RBAC • policy-as-code • immutable audit trail

HOW IT ADDRESSES THE PROBLEM

- Data, prompts, context and intermediate artifacts remain inside the organization-controlled boundary.
- Answers are evidence-linked: retrieve → cite → verify; low-confidence or policy-blocked tasks go to human review.
- Model-agnostic gateway lets the plant use approved open-weight LLM/VLMs without redesigning the workbench.

DESIGN DIFFERENTIATORS

Evidence-first RAG • Agent sandbox
Policy gate • Human-in-the-loop
Model routing • Audit-ready telemetry

KEYWORDS

SOVEREIGN AI | ZERO DATA EGRESS | OPEN-WEIGHT | MULTIMODAL RAG
HITL | POLICY GATE | MODEL AGNOSTIC | AUDITABILITY

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TECHNICAL APPROACH

Architecture & implementation flow

REQUEST → POLICY → AGENT → RETRIEVAL / TOOLS → MODEL → VERIFY → RESPONSE

USER / APP
Chat • upload • ask

→

POLICY GATE
RBAC • data classification
allow / deny / redact

→

AGENT ORCHESTRATOR
Plan • route • memory
state + tool permissions

→

MODEL GATEWAY
Model registry • routing
vLLM / Ollama

→

OPEN-WEIGHT LLM / VLM
Text + vision reasoning
local inference

KNOWLEDGE & DATA PLANE
Qdrant hybrid retrieval
Metadata filters: plant / unit / doc class
PostgreSQL: users • policies • audit metadata

TOOLS & CONNECTORS
PDF / image / OCR pipeline
REST • SQL • file-share adapters
Deterministic calculators & validators

SECURITY & OPERATIONS
On-prem Docker / Kubernetes
Secrets • network isolation • RBAC
Audit logs • model/tool telemetry

IMPLEMENTATION STACK

Python • FastAPI • React • LangGraph • vLLM / Ollama • Qdrant • PostgreSQL • OpenCV • PyMuPDF • Docker / Kubernetes

SECURITY PATTERN: Zero-Trust Boundary + Least-Privilege Tools + Human Approval for Operational Actions

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FEASIBILITY AND VIABILITY

Feasibility

- Software-first; designed for controlled on-prem compute.
- Open-weight serving + vector retrieval enable local deployment.
- Modular connectors/models allow phased rollout without platform rewrite.

Pilot validation plan

- Phase 1: document Q&A + maintenance knowledge search.
- Phase 2: visual inspection assistance + approved tools.
- Benchmark: factuality • citation coverage • latency • policy compliance.

CHALLENGES → MITIGATION

Hallucination / stale knowledge	Evidence-linked RAG + freshness metadata + verifier + abstain / review
Sensitive data exposure	On-prem / air-gapped mode + RBAC + policy gateway + audit logs
GPU / latency constraints	Model routing + quantization + batching + smaller task-specific models
Industrial integration	Adapter layer; start read-only, then controlled actions with approval

VIABILITY PRINCIPLE

BOUND THE SCOPE → MEASURE THE RISK → PROVE THE VALUE → SCALE WITH CONTROL

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IMPACT AND BENEFITS

Target users & operational value

ENGINEERING
SOP / drawing / specification Q&A

OPERATIONS
Shift knowledge • troubleshooting support

MAINTENANCE
Asset history • manuals • fault context

HSE / QUALITY
Evidence search • inspection assistance

BENEFITS

SECURITY & SOVEREIGNTY
Confidential industrial data stays within the controlled environment.

PRODUCTIVITY
One interface across documents, images, knowledge and approved tools.

TRACEABILITY
Evidence, tool/model events and user approvals can be logged for review.

SCALABILITY
Reusable connectors + model gateway support multiple plants / units / use cases.

SUCCESS MEASURES

- GROUNDED ANSWER RATE
- CITATION COVERAGE
- TASK COMPLETION TIME
- POLICY VIOLATIONS
- USER ADOPTION

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RESEARCH AND REFERENCES

Research & references supporting the architecture

Ollama	**Multimodal / local model execution** ollama.com/blog/multimodal-models
vLLM	**LLM inference and serving** docs.vllm.ai/en/stable/
Qdrant	**Hybrid retrieval, filtering, local vector search** qdrant.tech/documentation/
LangGraph	**Stateful agent workflows + human-in-the-loop** docs.langchain.com/oss/javascript/langgraph/thinking-in-langgraph
OpenCV	**Image processing + computer vision building blocks** docs.opencv.org/4.12.0/d1/dfb/intro.html

DESIGN NOTE

References support local serving, multimodal processing, retrieval and agent-workflow components. MRPL-specific connectors, data policies and performance targets should be validated during the pilot.

CORE DESIGN KEYWORDS: Sovereign AI • On-Prem • Air-Gapped • Open-Weight • Multimodal • Agentic RAG • HITL • Policy-as-Code • Auditability